

# Juhyun (Julie) Lee

juhyunle@asu.edu   
linkedin.com/in/julie-on-island   
julieonisland.github.io 

## Research Interests

**Deep Learning:** Multimodal Learning (Vision Language Model), Self-Supervised Learning, Agentic AI  
**AI for healthcare:** Computational Pathology, Medical Image Analysis

## Education

|                                                                                                  |                                             |
|--------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Arizona State University</b><br><i>Ph.D. in Computer Science (Advisor: Imon Banerjee)</i>     | AZ, United States<br>Aug. 2026 – May. 2030  |
| <b>University of North Carolina at Chapel Hill</b><br><i>Master's in Computer Science</i>        | NC, United States<br>Aug. 2024 – May. 2026  |
| <b>Chung-Ang University</b><br><i>B.S. in Applied Statistics &amp; B.A. in French Literature</i> | Seoul, South Korea<br>Mar. 2020 – Aug. 2023 |

## Publications

### **Structural MRI-Informed Multimodal Fusion for Robust Alzheimer's Disease Prediction**

**Juhyun Lee**, Sungjoon Park, Kyungwook Lee, Soorin Yim, Doyeong Hwang, Dongyun Kim, Kiwoong Yoo, Soonyoung Lee, Kiyoung Kim  
*NeurIPS Imageomics Workshop, 2025*

### **Proformer: a hybrid macaron transformer model predicts expression values from promoter sequences**

Il-Youp Kwak, Byeong-Chan Kim, **Juhyun Lee**, Taein Kang, Daniel J Garry, Jianyi Zhang, Wuming Gong  
*BMC Informatics, 2024*

## Work Experiences

|                                                                                                                                                                                                                                                                                                                       |                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>LinkedIn</b><br><i>AI/ML Engineer Intern</i>                                                                                                                                                                                                                                                                       | Mountain View, CA<br>May. 2026 – Aug. 2026  |
| <b>LG AI Research</b><br><i>Research Scientist Intern (Mentor: Sungjoon Park, Kiyoung Kim)</i>                                                                                                                                                                                                                        | Seoul, South Korea<br>May. 2025 – Aug. 2025 |
| <ul style="list-style-type: none"><li>Engineered a multimodal data processing pipeline and integrated it into a multimodal model, boosting Alzheimer's Disease classification accuracy by 20% over SOTA</li><li>Architected a modular Plan-Act-Memory AI agent to automate biomedical tasks using LangChain</li></ul> |                                             |

## Research Experiences

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>UNC Carmichael Lab</b><br><i>Research Assistant (Advisor: Iain Carmichael)</i>                                                                                                                                                                           | NC, United States<br>Jan. 2026 – May. 2026 |
| <ul style="list-style-type: none"><li>Achieved 0.7 AUC in smoking status prediction from bladder cancer whole slide images by leveraging pathology foundation models and weakly supervised learning</li></ul>                                               |                                            |
| <b>UNC ViVE Lab</b><br><i>Research Assistant (Advisor: Saif Khairat)</i>                                                                                                                                                                                    | NC, United States<br>Aug. 2025 – Dec. 2025 |
| <ul style="list-style-type: none"><li>Fine-tuned Llama-3-8B using LoRA (PEFT) for clinical reasoning; deployed with vLLM and Flask</li><li>Developed a speech-to-speech system for real-time interaction using OpenAI API, WebRTC, and TypeScript</li></ul> |                                            |
| <b>UNC Yap Lab</b><br><i>Research Assistant (Advisor: Pew-Thian Yap)</i>                                                                                                                                                                                    | NC, United States<br>Jan. 2025 – May. 2025 |

- Developed an Implicit Neural Representation (INR) framework that yielded 10x data compression while maintaining 92% reconstruction fidelity

#### UNC Data Driven EnviroLab

Research Assistant (Advisor: Angel Hsu)

NC, United States

Aug. 2024 – Dec. 2024

- Implemented a RAG chatbot using LangChain, FAISS, and LLM APIs to deliver accurate, relevant responses on climate policy

#### Chung-Ang University

Research Intern (Advisor: Il-young Kwak)

Seoul, South Korea

Mar. 2022 – June. 2023

## Conference Presentations

- **UNC Innovate Cancer Data Science Symposium** Jan. 2026  
*Poster Presentation*
- **NeurIPS Imageomics Workshop** Dec. 2025  
*Poster Presentation*
- **UNC Graduate Research Symposium** Oct. 2025  
*Poster Presentation*
- **UNC Data Science Day** Sep. 2025  
*Poster Presentation*
- **International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting** May 2025  
*Oral Presentation*

## Academic Services

#### Reviewer

ICML 2026, NeurIPS 2026

#### Mentorship

UNC Graduate Women in Computer Science Mentor

#### Teaching

- Instructor for *Medical Image Analysis and Computer Vision* for UNC PRIMES Summer School 2025

## Awards/Honors

- Summa Cum Laude**, Chung-Ang University Aug. 2023
- Random Promoter Dream Challenge 2022 (3rd place)** Sep. 2022
- Department Honor Scholarship**, Chung-Ang University Aug. 2020

## Skills

**Programming Languages:** Python, MATLAB, Java, JavaScript, SQL, R

**ML & Data Science:** Pytorch, TensorFlow, LangChain, Scikit-Learn, NumPy, Pandas, Matplotlib, SAS

**Web Development:** SpringBoot, Git, VueJS, AWS